LAMMPS official dem15-hooke-oblique

Oblique collision: friction impulse and spin | 29 October 2026

PASS	Four main runs evaluated using the unchanged official selected-time analytic criteria; source tag unvetted.

Official quantities, errors and acceptance limits

Quantity	Reference / limit	CPU dt	GPU dt	CPU dt/2	GPU dt/2
left vx (m/s)	-0.7 relative limit 0.01	-0.6988656 err 0.001621 PASS	-0.6988656 err 0.001621 PASS	-0.6992324 err 0.001097 PASS	-0.6992324 err 0.001097 PASS
left vy (m/s)	1.575 relative limit 0.01	1.572728 err 0.001442 PASS	1.572728 err 0.001442 PASS	1.573227 err 0.001126 PASS	1.573227 err 0.001126 PASS
left omega_z (rad/s)	-425 relative limit 0.01	-428.1136 err 0.007326 PASS	-428.1136 err 0.007326 PASS	-427.6156 err 0.006154 PASS	-427.6156 err 0.006154 PASS
right vx (m/s)	0.7 relative limit 0.01	0.6988656 err 0.001621 PASS	0.6988656 err 0.001621 PASS	0.6992324 err 0.001097 PASS	0.6992324 err 0.001097 PASS
right vy (m/s)	-1.575 relative limit 0.01	-1.572728 err 0.001442 PASS	-1.572728 err 0.001442 PASS	-1.573227 err 0.001126 PASS	-1.573227 err 0.001126 PASS
right omega_z (rad/s)	-425 relative limit 0.01	-428.1136 err 0.007326 PASS	-428.1136 err 0.007326 PASS	-427.6156 err 0.006154 PASS	-427.6156 err 0.006154 PASS

Check time=7e-05 s; original step 7000. Half dt uses twice the check step at the same time. Each cell shows the native value and its own absolute/relative error; a reference limit is not a same-code epsilon.

Original bodies and source-to-native mapping

Body	Radius / density	Initial xyz (m)	Initial v (m/s)	Initial spin (rad/s)
left	R=0.0025 rho=2700	(-0.00255, -0.0001, 0)	(1, 2, 0)	(0, 0, 0)
right	R=0.0025 rho=2700	(0.00255, 0.0001, 0)	(-1, -2, 0)	(0, 0, 0)

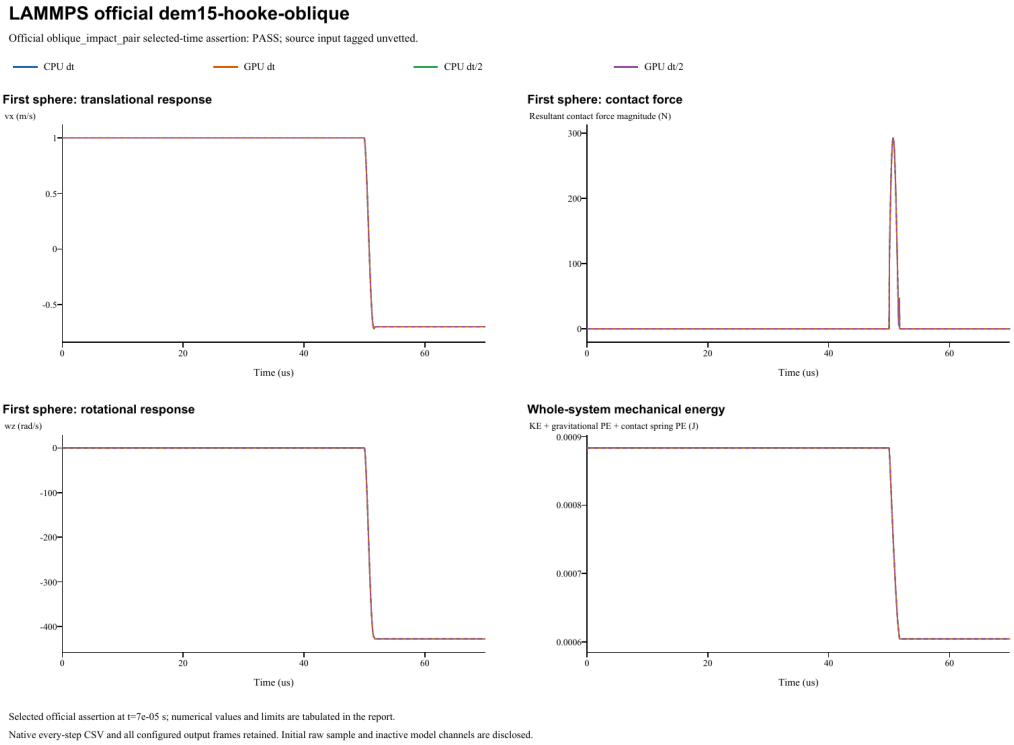
Linear Hooke Fn_elastic=kn*delta, kn=3e+08 N/m. Manual c_n=36.7327514 N s/m, c_t=0 N s/m; automatic restitution damping is inactive. Tangential inputs: kt=8e+07 N/m (explicit in linear mode), mu=0.25. Rolling/twist coefficients: kr=0, mu_r=0, ktwist=0, mu_twist=0. Source pair spin-slip uses R; native uses R-delta/2. Pair torque arms both use R-delta/2. Maximum source/native contact-speed difference is 0.000275732 m/s; torque-arm difference is 8.47033e-22 m. Official selected-time acceptance does not imply pointwise kinematic identity.

Gravity=(0, 0, 0) m/s^2. Base/half dt=1e-08/5e-09 s; 7000/14000 steps. Exact constant normal restitution dashpot; source tangential damping ratio is0, so manual ct=0. Full finite partner masses, oblique impact parameters and spin generation retained.

Pinned LAMMPS source

Official input | Analytic checker | Parameter mapping.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; input blob 32b2385c6f9d69c3d3d0af7bdb7dd0cd45cef81b. Official tag: unvetted.
LAMMPS was not rerun locally. Physical inputs were not fitted to native results.

LAMMPS native histories and refinement



CPU/GPU and half-step comparison

Quantity	CPU dt error	CPU dt/2 error	Error type / limit
left vx (m/s)	0.0016206	0.0010966	relative: 0.01
left vy (m/s)	0.0014425	0.0011256	relative: 0.01
left omega_z (rad/s)	0.0073262	0.0061544	relative: 0.01
right vx (m/s)	0.0016206	0.0010966	relative: 0.01
right vy (m/s)	0.0014425	0.0011256	relative: 0.01
right omega_z (rad/s)	0.0073262	0.0061544	relative: 0.01

CPU/GPU whole-history maximum position / velocity / spin / force differences: 6.78e-21 m / 4.85e-14 m/s / 1.12e-11 rad/s / 2.64e-10 N.

Maximum individual impulse / position-drift / velocity-kick residuals: 3.25e-19 kg m/s / 2.17e-19 m / 2.15e-16 m/s. Gravity is added separately to saved native contact forces.

Complete results and verified scope

All 2,804 native frames are retained; main frame counts=[701, 701, 701, 701]. Every numerical-step particle/contact CSV, four loadable main projects and raw initial samples remain preserved. Output cadence is recorded separately from integration dt. Energy/history diagnostics do not replace the official acceptance quantities.

Run manifest | Analysis | Official metrics | Independent audit | Full native ZIP | Frame hashes

Source checkpoints: Source checkpoint states are supplemental regression diagnostics. Their differences are reported directly; the YAML same-code epsilon is not imposed as an independent cross-software acceptance limit.

A stable helper executable is statically linked to the current native CPU/CUDA libraries. Historical GUI/library byte identity is not asserted. Scope is the active source physics and the selected official criteria; inactive options remain outside this test.